

ITA0605 - Machine Learning

List of Lab Exercises

12.Implement Iris Flower Classification using KNN.

AIM

To implement Iris Flower Classification using the K-Nearest Neighbors (KNN) machine learning algorithm and classify iris flowers into their respective species based on sepal and petal measurements.

ALGORITHM

1. Import the required Python libraries.
2. Load the built-in Iris dataset.
3. Separate the input features and target classes.
4. Split the dataset into training and testing sets.
5. Standardize the feature values using `StandardScaler`.
6. Create a KNN classifier with a suitable value of K.
7. Train the KNN model using the training dataset.
8. Predict the species of flowers in the test dataset.
9. Calculate the classification accuracy.
10. Display the actual and predicted classes.

PROGRAM

DATA SET 1:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
X = [
    [150, 7.5, 90],
    [160, 7.8, 88],
```

```

[145, 7.3, 91],
[120, 5.5, 70],
[125, 5.8, 72],
[130, 6.0, 74],
[155, 7.6, 89],
[118, 5.4, 69],
[148, 7.4, 92],
[122, 5.7, 71]
]
y = [
    "Apple", "Apple", "Apple", "Orange", "Orange",
    "Orange", "Apple", "Orange", "Apple", "Orange"
]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
new_sample = [[152, 7.5, 90]]
prediction = knn.predict(new_sample)
print("Predicted Class:", prediction[0])

```

OUTPUT:

```

▶ from sklearn.neighbors import KNeighborsClassifier
  from sklearn.model_selection import train_test_split
  X = [
      [150, 7.5, 90],
      [160, 7.8, 88],
      [145, 7.3, 91],
      [120, 5.5, 70],
      [125, 5.8, 72],
      [130, 6.0, 74],
      [155, 7.6, 89],
      [118, 5.4, 69],
      [148, 7.4, 92],
      [122, 5.7, 71]
  ]
  y = [
      "Apple", "Apple", "Apple", "Orange", "Orange",
      "Orange", "Apple", "Orange", "Apple", "Orange"
  ]
  X_train, X_test, y_train, y_test = train_test_split(
      X, y, test_size=0.3, random_state=42
  )
  knn = KNeighborsClassifier(n_neighbors=3)
  knn.fit(X_train, y_train)
  new_sample = [[152, 7.5, 90]]
  prediction = knn.predict(new_sample)
  print("Predicted Class:", prediction[0])

```

... Predicted Class: Apple

DATA SET 2:

```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
X = [
    [9.2,90,88],
    [8.8,85,82],
    [8.5,80,79],
    [7.2,72,70],
    [6.8,68,65],
    [9.0,92,90],
    [7.5,75,74],
    [8.6,84,81],

```

```

[6.5,60,58],
[8.9,88,86]
]
y = [
    "Placed", "Placed", "Placed", "Not Placed", "Not Placed",
    "Placed", "Not Placed", "Placed", "Not Placed", "Placed"
]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
new_sample = [[8.7,86,84]]
prediction = knn.predict(new_sample)
print("Predicted Class:", prediction[0])

```

OUTPUT:

```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
X = [
    [9.2,90,88],
    [8.8,85,82],
    [8.5,80,79],
    [7.2,72,70],
    [6.8,68,65],
    [9.0,92,90],
    [7.5,75,74],
    [8.6,84,81],
    [6.5,60,58],
    [8.9,88,86]
]
y = [
    "Placed", "Placed", "Placed", "Not Placed", "Not Placed",
    "Placed", "Not Placed", "Placed", "Not Placed", "Placed"
]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
new_sample = [[8.7,86,84]]
prediction = knn.predict(new_sample)
print("Predicted Class:", prediction[0])

```

... Predicted Class: Placed

DATA SET 3:

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.model_selection import train_test_split
```

```
X = [
```

```
    [39.0,110,94],
```

```
    [38.5,108,95],
```

```
    [37.0,78,99],
```

```
    [39.2,115,93],
```

```
    [36.8,75,98],
```

```
    [38.8,112,94],
```

```
    [37.2,80,98],
```

```
    [39.1,114,92],
```

```
    [36.7,74,99],
```

```
    [38.9,111,93]
```

```
]
```

```
y = [
```

```
    "Positive", "Positive", "Negative", "Positive", "Negative",
```

```
    "Positive", "Negative", "Positive", "Negative", "Positive"
```

```
]
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42
```

)

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train, y_train)
```

```
new_sample = [[38.7,109,94]]
```

```
prediction = knn.predict(new_sample)
```

```
print("Predicted Class:", prediction[0])
```

OUTPUT:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
X = [
    [39.0,110,94],
    [38.5,108,95],
    [37.0,78,99],
    [39.2,115,93],
    [36.8,75,98],
    [38.8,112,94],
    [37.2,80,98],
    [39.1,114,92],
    [36.7,74,99],
    [38.9,111,93]
]
y = [
    "Positive","Positive","Negative","Positive","Negative",
    "Positive","Negative","Positive","Negative","Positive"
]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
new_sample = [[38.7,109,94]]
prediction = knn.predict(new_sample)
print("Predicted Class:", prediction[0])
```

```
... Predicted Class: Positive
```

DATA SET 4:

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.model_selection import train_test_split
```

```
X = [
```

```
    [10,95,50],
```

```
    [8,90,45],
```

```
    [7,88,40],
```

```
    [3,65,20],
```

```
    [2,60,18],
```

```
    [9,93,48],
```

```
    [4,70,25],
```

```
    [8,89,42],
```

```
    [2,58,15],
```

```
    [6,85,38]
```

```
]
```

```
y = [
```

```
    "Promoted","Promoted","Promoted","Not Promoted",
```

```
    "Not Promoted","Promoted","Not Promoted",
```

```
    "Promoted","Not Promoted","Promoted"
```

```
]
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```

X, y, test_size=0.3, random_state=42

)

knn = KNeighborsClassifier(n_neighbors=3)

knn.fit(X_train, y_train)

new_sample = [[7,90,44]]

prediction = knn.predict(new_sample)

print("Predicted Class:", prediction[0])

```

OUTPUT:

```

▶ from sklearn.neighbors import KNeighborsClassifier
  from sklearn.model_selection import train_test_split
  X = [
    [10,95,50],
    [8,90,45],
    [7,88,40],
    [3,65,20],
    [2,60,18],
    [9,93,48],
    [4,70,25],
    [8,89,42],
    [2,58,15],
    [6,85,38]
  ]
  y = [
    "Promoted", "Promoted", "Promoted", "Not Promoted",
    "Not Promoted", "Promoted", "Not Promoted",
    "Promoted", "Not Promoted", "Promoted"
  ]

  X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
  )
  knn = KNeighborsClassifier(n_neighbors=3)
  knn.fit(X_train, y_train)
  new_sample = [[7,90,44]]
  prediction = knn.predict(new_sample)
  print("Predicted Class:", prediction[0])

... Predicted Class: Promoted

```

DATA SET 5:


```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.model_selection import train_test_split
```

```
X = [
```

```
    [100,95,70],
```

```
    [125,110,65],
```

```
    [150,125,55],
```

```
    [800,850,22],
```

```
    [1000,950,20],
```

```
    [1200,1050,18],
```

```
    [110,100,68],
```

```
    [900,900,21],
```

```
    [140,120,58],
```

```
    [1300,1100,17]
```

```
]
```

```
y = [
```

```
    "Bike","Bike","Bike","Car","Car",
```

```
    "Car","Bike","Car","Bike","Car"
```

```
]
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42
```

```
)
```

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train, y_train)
```

```
new_sample = [[135,118,60]]
```

```
prediction = knn.predict(new_sample)
```

```
print("Predicted Class:", prediction[0])
```

OUTPUT:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
X = [
    [100,95,70],
    [125,110,65],
    [150,125,55],
    [800,850,22],
    [1000,950,20],
    [1200,1050,18],
    [110,100,68],
    [900,900,21],
    [140,120,58],
    [1300,1100,17]
]
y = [
    "Bike","Bike","Bike","Car","Car",
    "Car","Bike","Car","Bike","Car"
]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
new_sample = [[135,118,60]]
prediction = knn.predict(new_sample)
print("Predicted Class:", prediction[0])
```

```
... Predicted Class: Bike
```

RESULT

The Iris Flower Classification was successfully implemented using the K-Nearest Neighbors (KNN) algorithm. The model classifies iris flowers into Setosa, Versicolor, and Virginica based on sepal and petal measurements. The classification accuracy and actual versus predicted classes are displayed as output.